

Answer Key — Biology Grade 11

Sub-Strand 3.2: Growth and Development in Animals • Lesson 5: The Chemical Switch: How Juvenile Hormone Decides Whether an Insect Stays Young or Becomes an Adult

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	B	D	A	C	C	C

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share prediction of what inside the insect tells it to moult

Q1. A learner says: "A caterpillar just melts inside the pupa and rebuilds itself without any control." Which fact best challenges this idea?

- A. Caterpillars of one species always become the same kind of butterfly, which suggests the changes are controlled by chemical signals in the body**
- B. Caterpillars shed their skin, and the shed skin controls the next stage
- C. Pupae are always found on maize leaves, so the leaf controls the change
- D. Caterpillars stop eating when they become pupae, so nothing is happening inside

Correct answer: A. A species always makes the same adult form, so the process is regulated. Hormones such as ecdysone and JH act as the control signals.

USE AFTER: OBSERVE PHASE — Observe: groups view animations and fill the observation table (Hormone(s) present, Level of JH, Result of moult)

Q2. Which statement correctly describes the roles of the two hormones in insect growth?

- A. Ecdysone and JH both cause moulting, but neither affects the stage that follows
- B. Ecdysone triggers moulting and the level of JH decides the type of moult**
- C. JH triggers moulting and ecdysone decides the type of moult
- D. JH is released by the prothoracic gland and it makes the insect stop moulting forever

Correct answer: B. Ecdysone, from the prothoracic gland, triggers each moult. The level of JH, from the corpora allata, decides whether the result is another nymph or larva or a pupa or adult.

USE AFTER: OBSERVE PHASE — Observe: pause points in the animation asking 'What just happened to the JH level?'

Q3. A locust nymph in Turkana is about to moult. At this moult its Juvenile Hormone (JH) level is high. What will most likely result from the moult?

- A. A pupa
- B. No moult will take place because JH stops moulting
- C. A fully grown adult locust
- D. Another, larger nymph**

Correct answer: D. High JH keeps the juvenile features. Ecdysone still triggers the moult, but the result is another larger nymph.

USE AFTER: MODEL BUILDING PHASE — Model: exit ticket question 2 and the revised model with the high-JH/low-JH pathways

Q4. A butterfly caterpillar has finished growing and its JH level starts to fall. What does the falling JH cause at the next moults?

- A. The caterpillar becomes a pupa, and later an adult butterfly**
- B. The caterpillar changes straight into an adult without a pupa
- C. The caterpillar stops producing ecdysone and never moults again
- D. The caterpillar moults into a bigger caterpillar again and again

Correct answer: A. In complete metamorphosis, falling JH makes the larva moult into a pupa. A further moult then gives the adult butterfly.

USE AFTER: EXPLAIN PHASE — Explain: flow diagram pathways (B) for incomplete and complete metamorphosis, and the question on why grasshoppers have no pupa stage

Q5. Why does a grasshopper have no pupa stage?

- A. Grasshopper nymphs have no exoskeleton, so they do not need a pupa
- B. Grasshoppers keep their JH level high for their whole life
- C. It has incomplete metamorphosis: when JH falls, the last nymph moults directly into an adult**
- D. Grasshoppers do not make ecdysone, so they cannot moult into a pupa

Correct answer: C. Grasshoppers have incomplete metamorphosis. The nymphs look like small adults, and when JH falls the final nymph moults straight into an adult without a pupa stage.

USE AFTER: EXPLAIN PHASE — Explain: CER teacher probe 'What would you expect if JH were artificially added to a final-instar caterpillar?'

Q6. Scientists add extra JH to a caterpillar in its final larval stage, when JH would normally fall. What is the most likely result?

- A. It becomes an adult butterfly at once
- B. It becomes a pupa earlier than normal
- C. It makes an extra larval stage, or a very large larva, instead of becoming a pupa**
- D. Nothing changes, because only ecdysone affects the type of moult

Correct answer: C. High JH keeps the larval form. Adding JH when it should be falling delays the change to a pupa, giving an extra larval stage or a giant larva.

USE AFTER: EXPLAIN PHASE — Explain: using the diagram to explain the exuvia in the school garden and why the new grasshopper is larger than the discarded skin

Q7. A newly emerged grasshopper is larger than the empty exuvia (old skin) it left behind. Which explanation is correct?

- A. The hard exoskeleton stretches smoothly as the grasshopper grows
- B. The old skin peeled off like sunburnt skin and the grasshopper grew bigger afterwards
- C. The grasshopper's body expands before its new cuticle hardens**
- D. The old skin shrank after it was shed

Correct answer: C. After shedding the old exoskeleton, the insect expands its body while the new cuticle is still soft. The cuticle then hardens, so the insect is larger than the old skin and growth happens in steps.